Gradient Descent Dynamics of 1-Hidden Layer DNN

1 Reformulation of Gradient Descent in Generalized Fourier Series (GFS)

1.1 The 2-norm Loss and Derived Generalized Fourier Series Expansion

The problem setup is to fit J given pairs of feature and label (inputs and outputs) $\{(x_j, y_j)\}$ by a function $\Upsilon_\theta(x)$ with parameter θ (usually a vector). Without loss of generality, we limit the range of x_j to $[-1, 1]$. Generalize to high dimensional $x_j \in [-1, 1]^d$ is under consideration.

We have the 2-norm loss for the error of the fitting:

$$\tilde{L}_J(\theta) = \frac{1}{2J} \sum_{j=1}^{J} \left(\Upsilon_\theta(x_j) - y_j\right)^2 . \tag{1}$$

Assume the samples $\{x_j\}$ are draw from a distribution $\rho(x)$, and the labels are come from an underlying function $y(x)$: $y_j = y(x_j)$. Now let $J \to \infty$, we should have a theoretical loss $L(\theta)$:

$$L(\theta) = \frac{1}{2} \int_\Omega \left(\Upsilon_\theta(x) - y(x)\right)^2 \rho(x) \, \mathrm{d}x. \tag{2}$$

$$\tilde{L}_J(\theta) \to L(\theta) \quad \text{as } J \to \infty,$$

The Eq. (2) naturally induced a norm and an inner product (assume all real functions)

$$\|f\|_\rho = \int_\Omega f^2(x)\rho(x) \, \mathrm{d}x, \tag{3}$$

$$\langle f, g \rangle_\rho = \int_\Omega f(x)g(x)\rho(x) \, \mathrm{d}x. \tag{4}$$

Under the inner product in Eq. (4), there exist a set of orthonomal polynomial basis $\{p_k(x)\}_{k=0}^\infty$ where $p_k(x)$ is a polynomial of order k:

$$\langle p_k, p_j \rangle_\rho = \delta_{kj}. \tag{5}$$

In 1-D case, each $p_k(x)$ has exactly k zeros. Therefore the subscript k has similar meaning as the k in Fourier mode $e^{\mathrm{i}kx}$, i.e. the frequency.

If $\rho(x) > 0$ almost everywhere, then $\{p_k(x)\}_{k=0}^\infty$ is a complete basis [1] with the same convergence trends as Chebyshev approximation. [2]

[1] TODO: ref

[2] TODO: ref

The fitting function Υ_θ, assumed piecewise continuous, can be expanded in Generalized Fourier Series (GFS):

$$\Upsilon_\theta(x) = \sum_{k=0}^{\infty} c_k(\theta) p_k(x) \quad \text{(a.e.)} \tag{6}$$

$$c_k(\theta) = \langle \Upsilon_\theta, p_k \rangle_\rho \tag{7}$$

where (a.e.) here means (equal) almost everywhere. If $\Upsilon_\theta(x)$ is continuous, then it is pointwise equal. In later context we will denote both cases by simply equal "=".

1.2 Effectiveness of GFS in Representing Smooth Functions

See Ref. [1, 2]. Here list only theorems for Chebyshev polynomials.

1.2.1 Chebyshev Approximations

Theorems for absolutely continuous functions.

Lemma 1 (Theorem 7.1 in [2]). *Assume $f, \ldots, f^{(\nu-1)}$ be absolutely continuous on $[-1, 1]$ and $f^{(\nu)}$ is of bounded variation V. Then for $\forall k \geq \nu + 1$, the Chebyshev coefficients of f satisfy*

$$|a_k| \leq \frac{2V}{\pi k(k-1)\cdots(k-\nu)} \leq \frac{2V}{\pi(k-\nu)^{\nu+1}}.$$

Lemma 2 (Theorem 7.2 in [2]). *The same condition as Lemma 1, then for $\forall n > \nu$, its Chebyshev projections satisfy*

$$\|f - f_n\|_\infty \leq \frac{2V}{\pi\nu(n-\nu)^\nu}$$

and its Chebyshev interpolants satisfy

$$\|f - p_n\|_\infty \leq \frac{4V}{\pi\nu(n-\nu)^\nu}.$$

Theorems for analytical functions.

Lemma 3 (Theorem 8.1 in [2]). *Assume f analytic in $[-1, 1]$ be analytically continuable to the open Bernstein ellipse E_ρ (note [3]), where it satisfies $|f(x)| \leq M$ for some M. Then its Chebyshev coefficients satisfy $|a_0| \leq M$ and*

$$|a_k| \leq 2M\rho^{-k}, \quad k \geq 1.$$

Lemma 4 (Theorem 8.2 in [2]). *The same condition as Lemma 3, then for $\forall n \geq 0$ its Chebyshev projections satisfy*

$$\|f - f_n\|_\infty \leq \frac{2M\rho^{-n}}{\rho - 1}$$

and its Chebyshev interpolants satisfy

$$\|f - p_n\|_\infty \leq \frac{4M\rho^{-n}}{\rho - 1}.$$

[3] Bernstein ellipse E_ρ: foci 1 and -1 with semimajor and semi-minor axis lengths summing to ρ in complex plane.

1.3 Evolution of GFS in Gradient Descent

The evolution of parameter θ depends on the gradient of loss. In terms of GFS, the gradient can be written as:

$$\begin{aligned}\frac{\partial L(\theta)}{\partial\theta} &= \frac{1}{2}\frac{\partial}{\partial\theta}\left\langle \Upsilon_\theta - y, \Upsilon_\theta - y\right\rangle_\rho = \frac{1}{2}\frac{\partial}{\partial\theta}\sum_{k=0}^{\infty}\left\langle \Upsilon_\theta - y, p_k\right\rangle_\rho^2 = \sum_{k=0}^{\infty}\left\langle \Upsilon_\theta - y, p_k\right\rangle_\rho \frac{\partial}{\partial\theta}\left\langle \Upsilon_\theta, p_k\right\rangle_\rho \\ &= \sum_{k=0}^{\infty}\left(c_k(\theta) - \hat{y}_k\right)\frac{\partial c_k(\theta)}{\partial\theta}, \end{aligned} \tag{8}$$

where assuming $y(x)$ is piecewise continuous (at least L^2) and

$$\hat{y}_k \triangleq \left\langle y, p_k\right\rangle_\rho \quad k = 0, 1, \ldots \tag{9}$$

$$\frac{\partial c_k(\theta)}{\partial\theta} = \left\langle \frac{\partial \Upsilon_\theta}{\partial\theta}, p_k\right\rangle_\rho . \tag{10}$$

The trajectory of $\theta(t)$ in gradient descent (GD) algorithm can be rewritten in ODE as a limit of small step size:

$$\frac{\mathrm{d}\theta}{\mathrm{d}t} = -\alpha\left[\frac{\partial L(\theta)}{\partial\theta}\right]^T, \tag{11}$$

where α is the learning rate, $\cdot^T$ means transpose of the Jacobian matrix inside, and

$$\left[\frac{\partial L(\theta)}{\partial\theta}\right] \triangleq \begin{bmatrix}\frac{\partial L(\theta)}{\partial\theta_1} & \frac{\partial L(\theta)}{\partial\theta_2} & \ldots\end{bmatrix}.$$

The dynamics of the GFS coefficients $c_k(\theta(t))$ can be derived with help of Eq. (8) and Eq. (11):

$$\begin{aligned}\frac{\mathrm{d}(c_k(\theta(t)) - \hat{y}_k)}{\mathrm{d}t} &= \frac{\partial c_k(\theta)}{\partial\theta}\cdot\frac{\mathrm{d}\theta}{\mathrm{d}t} \\ &= -\alpha\sum_{k'=0}^{\infty}\left[\frac{\partial c_k(\theta)}{\partial\theta}\right]\left[\frac{\partial c_{k'}(\theta)}{\partial\theta}\right]^T\left(c_{k'}(\theta) - \hat{y}_{k'}\right), \quad k = 0, 1, \ldots. \end{aligned} \tag{12}$$

Note that, formally, a system of linear equation

$$\frac{\mathrm{d}\vec{r}}{\mathrm{d}t} = -\alpha H\vec{r}, \tag{13}$$

with constant symmetric matrix H and spectral decomposition $H = P\Lambda P^{-1}$ has solution:

$$\vec{r}(t) = e^{-\alpha H t}\vec{C} = Pe^{-\alpha\Lambda t}P^{-1}\vec{C}, \tag{14}$$

where $\vec{C}$ is an arbitrary constant vector.

It is clear that the eigenvalues of H govern the dynamics of Eq.(13).

With substitution

$$\vec{r}(t) = \left[c_k(\theta(t)) - \hat{y}_k\right]_{k\in\mathbb{N}_0}, \tag{15}$$

$$H = H_{c(\theta)} \triangleq \left[\frac{\partial c_k(\theta)}{\partial \theta}\right]_{k\in\mathbb{N}_0} \left[\frac{\partial c_k(\theta)}{\partial \theta}\right]_{k\in\mathbb{N}_0}^T, \tag{16}$$

where $[\cdot]_{k\in\mathbb{N}_0}$ means vertically stack the terms indexed by k to form a matrix. e.g.

$$\left[\frac{\partial c_k(\theta)}{\partial \theta}\right]_{k\in\mathbb{N}_0} \triangleq \begin{bmatrix} \frac{\partial c_0(\theta)}{\partial \theta_1} & \frac{\partial c_0(\theta)}{\partial \theta_2} & \cdots \\ \frac{\partial c_1(\theta)}{\partial \theta_1} & \frac{\partial c_1(\theta)}{\partial \theta_2} & \\ \vdots & & \ddots \end{bmatrix}.$$

We see the study of eigenvalues could greatly helps the understanding of Eq.(12), despite $H_{c(\theta)}$ is now time dependent. Now we turn to the analysis of matrix $H_{c(\theta)}$.

2 Structure of the matrix H

H is an $\infty \times \infty$ matrix, for simplicity, assume it is finitely truncated. The truncation is reasonable due to the fast decaying property of $c_k(\theta)$. Analysis of the infinite dimension problem directly is under consideration.

Now we need to know some details of $\Upsilon_\theta(x)$ for the investigation of matrix H.

2.1 1-Hidden Layer DNN

The 1-Hidden Layer DNN with parameter $\theta = \{a_l, w_l, b_l\}_{l=1}^L$ and input range $x \in [-1, 1]$ is:

$$\Upsilon_\theta(x) \triangleq \sum_{l=1}^{L} a_l \sigma(w_l x + b_l). \tag{17}$$

Due to (10), matrix H of Eq.(16) is computed from derivatives of Υ_θ :

$$\frac{\partial \Upsilon_\theta(x)}{\partial \theta} = \begin{cases} \sigma(w_l x + b_l) & \theta = a_l \\ a_l \sigma'(w_l x + b_l) & \theta = b_l \\ a_l \sigma'(w_l x + b_l) x & \theta = w_l \end{cases} \quad l = 1, \ldots, L. \tag{18}$$

The smoothness of $\sigma(x)$ determines the smoothness of $\frac{\partial \Upsilon_\theta}{\partial \theta}$ and hence the decaying speed of $\frac{\partial c_k(\theta)}{\partial \theta}$ along k. Notably there are two common cases:

Case 1 $\sigma(x)$ is analytical in a neighborhood of $[-1, 1]$ in complex plane (in the sense of Lemma 3), so $\Upsilon_\theta(x)$ and $\frac{\partial \Upsilon_\theta(x)}{\partial \theta}$ are analytical also. Let's further assume that there is no singularity in a neighborhood of the real axis [4] for the domain of $\sigma(x)$ in complex plane. i.e. the nearest singularity is $\pm \mathrm{i}B$ $(B > 0)$ away from the real axis. For example, $\sigma(x) = \mathrm{sigmoid}(x) = 1/(1 + \exp(-x))$, has singularity $x = \mathrm{i}(\pi + 2n\pi)$ $\forall n \in \mathbb{Z}$, thus $B = \pi$.

In this setup, the Bernstein ellipse parameter [5] is $\Theta_{\sigma(x)} \geq B + \sqrt{1 + B^2}$.

[4] If there is singularity on the real axis, then it will be non-smooth with certain translation parameter b_l. The estimation of Θ will be more complicated and problem dependent.

[5] The Θ is the ρ in lemma 3, not to be confused with the weighting $\rho(x)$.

So the Bernstein ellipse parameter for $\frac{\partial \Upsilon_\theta}{\partial \theta}$ is $\Theta_{\sigma(w_l x)} = B/w_l + \sqrt{1 + (B/w_l)^2}$, we get

$$\left|\frac{\partial c_k(\theta)}{\partial \theta}\right| < 2M\Theta_{\sigma(w_l x)}^{-k} = 2M\left(B/w_l + \sqrt{1 + (B/w_l)^2}\right)^{-k}, \quad k \geq 1 \tag{19}$$

$$\approx 2M\left(\Theta_{\sigma(x)}/w_l\right)^{-k} \quad (\text{when } w_l \ll B) \tag{20}$$

Case 2 $\sigma(x)$ is continuous up to $(\nu-1)$-th derivative (in the sense of Lemma 1), so $\Upsilon_\theta(x)$ also. And $\frac{\partial \Upsilon_\theta(x)}{\partial \theta}$ are continuous up to $(\tilde{\nu}-1)$-th derivative, with $\tilde{\nu} = \nu - 1$ for $\theta = b_l$ and w_l, or $\tilde{\nu} = \nu$ for $\theta = a_l$. By lemma 1, we have upper bound

$$\left|\frac{\partial c_k(\theta)}{\partial \theta}\right| \leq \frac{2V}{\pi(k-\tilde{\nu})^{\tilde{\nu}+1}} \quad (\forall k > \tilde{\nu}). \tag{21}$$

2.2 Graded Matrix

Both Eq.(19) and Eq.(21) show the decay of $\left|\frac{\partial c_k(\theta)}{\partial \theta}\right|$ with respect to k. Thus the matrix Eq.(16) will also have certain decaying property. Specifically, there exist symmetric matrix B of bounded spectral norm and diagonal matrix D such that

$$H = DBD \tag{22}$$

where matrix D can be defined as $D = \mathrm{diag}(\left[\Theta_{\sigma(\max(w_l)x)}^{-k}\right]_{k\in\mathbb{N}_0})$ or $D = \mathrm{diag}(\left[(k+1)^{\min(v)-1}\right]_{k\in\mathbb{N}_0})$, depending on the smoothness of $\sigma(x)$. Or, a more straightforward way is to define $D = \mathrm{diag}([H_{kk}]_{k\in\mathbb{N}})$. And $B = D^{-1}HD^{-1}$ where pseudo-inverse of D is used if necessary. In the later definition of D, it is easy to show that $D_{kk} = O(\Theta_{\sigma(\max(w_l)x)}^{-k})$ or $D_{kk} = O(k^{-\min(\tilde{\nu})-1})$. The matrix H with structure Eq.(22) is called graded matrix.

For matrix that has structure Eq.(22), due to Min-max theorem of eigenvalues, we have

$$\lambda_k(H) \leq D_{kk}^2 \lambda_{\max}(B) \quad \forall k \in \mathbb{N} \tag{23}$$

where $\lambda_k(H)$ is the k-th eigenvalue of H sorting in descending order. Due to $H_{jj} = D_{kk}^2 B_{jj}$, Eq. (23) can also written as (assume $B_{jj} \neq 0$)

$$\lambda_k(H) \leq H_{jj}\frac{\lambda_{\max}(B)}{B_{jj}} \quad \forall k \in \mathbb{N}. \tag{24}$$

If $\lambda_{\min}(B) > C > 0$, then the eigenvectors of H are close to unit vector.

TODO: add pictures.

Another way to derive it is by the Schur–Horn theorem. The sorted array $\left[H_{r(j)r(j)}\right]_{j\in\mathbb{N}_0}$ of diagonal $[H_{jj}]_{j\in\mathbb{N}_0}$ satisfies

$$\sum_{j=1}^{k} H_{r(j)r(j)} \leq \sum_{j=1}^{k} \lambda_j \quad k \in \mathbb{N} \tag{25}$$

and

$$\sum_{j=1}^{\infty} H_{r(j)r(j)} = \sum_{j=1}^{\infty} \lambda_j \tag{26}$$

Eq. (26) minus Eq. (25), with the fact that $\lambda_j \geq 0$, $H_{jj} \geq 0$, we get

$$\lambda_k \leq \sum_{j=k}^{\infty} \lambda_j \leq \sum_{j=k}^{\infty} H_{r(j)r(j)}. \tag{27}$$

For example, if $\left|\frac{\partial c_k(\theta)}{\partial \theta_j}\right| \leq C_0 \Theta^{-k}\ \forall k, j$,

$$|H_{jk}| \leq C_0^2 L \Theta^{-j-k} \tag{28}$$

$$\lambda_k \leq C_0^2 L \frac{\Theta^{-2k}}{1-\Theta^{-2}} \tag{29}$$

A tighter bound can be obtained by (not proved "$\leq$")

$$\lambda_k(H) = \lambda_k(RDDR^T) \leq C \cdot \left[RDDR^T\right]_{kk} \tag{30}$$

where R comes from QR decomposition

$$QR = \left[\frac{\partial c_k(\theta)}{\partial \theta}\right]_{k \in \mathbb{N}_0}^T D^{-1}. \tag{31}$$

Or alternatively by Cholesky decomposition (TODO verify this)

$$R^T R = B \tag{32}$$

Further properties:

$$\left[RDDR^T\right]_{kk} = \sum_{j=k}^{\infty} R_{kj}^2 D_{jj}^2. \tag{33}$$

$$\left|\left[DR^T\right]_{ij}\right| \leq \begin{cases} C_1 D_{ii} & i \geq j \\ 0 & i < j \end{cases}$$

$$\left|\left[RDDR^T\right]_{ij}\right| \leq \sum_{l=\max\{i,j\}} C_1^2 D_{ll}^2$$

The matrix $RDDR^T$ has better decaying structure compare to H

$$\left|\left[RDDR^T\right]_{ij}\right| \leq C_1^2 \frac{\Theta^{-2\max\{i,j\}}}{1-\Theta^{-2}}. \tag{34}$$

3 Related Problems That Encounter Similar Matrix H

3.1 Hessian of Loss Function

The hessian of loss function is

$$\frac{\partial L(\theta)}{\partial \theta} = \int_\Omega (\Upsilon_\theta(x) - y(x)) \frac{\partial \Upsilon_\theta(x)}{\partial \theta} \rho(x)\, \mathrm{d}x \tag{35}$$

$$\nabla_\theta \otimes \nabla_\theta L(\theta) = \int_\Omega \left(\left[\frac{\partial \Upsilon_\theta(x)}{\partial \theta}\right]^T \cdot \left[\frac{\partial \Upsilon_\theta(x)}{\partial \theta}\right] + (\Upsilon_\theta(x) - y(x)) \nabla_\theta \otimes \nabla_\theta \Upsilon_\theta(x) \right) \rho(x)\, \mathrm{d}x \tag{36}$$

Define

$$H_{L(\theta)} \triangleq \int_{\Omega} \left[\frac{\partial \Upsilon_\theta(x)}{\partial \theta}\right]^T \cdot \left[\frac{\partial \Upsilon_\theta(x)}{\partial \theta}\right] \rho(x)\,\mathrm{d}x. \tag{37}$$

When the θ reaches the global optimal solution θ^* such that $\Upsilon_{\theta^*}(x) \equiv y(x)$, the hessian become

$$\nabla_\theta \otimes \nabla_\theta L(\theta^*) = H_{L(\theta^*)}. \tag{38}$$

When writing in terms of GFS,

$$\left[H_{L(\theta)}\right]_{ij} = \left\langle \frac{\partial \Upsilon_\theta(x)}{\partial \theta_i}, \frac{\partial \Upsilon_\theta(x)}{\partial \theta_j} \right\rangle_\rho = \sum_{k=0}^{\infty} \frac{\partial c_k(\theta)}{\partial \theta_i} \cdot \frac{\partial c_k(\theta)}{\partial \theta_j},$$

we have

$$H_{L(\theta)} = \left[\frac{\partial c_k(\theta)}{\partial \theta}\right]_{k\in\mathbb{N}_0}^T \left[\frac{\partial c_k(\theta)}{\partial \theta}\right]_{k\in\mathbb{N}_0}. \tag{39}$$

The matrix $H_{L(\theta)}$ and $H_{c(\theta)}$ (Eq.(16)) are differed only by the multiplication order of the transposed matrix, therefore they have exactly the same eigenvalues. For simplicity, define $\lambda_k = 0$ when k exceed corresponding matrix dimension.

$$\lambda_k(H_{L(\theta)}) = \lambda_k(H_{c(\theta)}) \quad (\forall k \in \mathbb{N}) \tag{40}$$

Thus from Sec. 2.2 we know that the landscape of loss is extremely flat due to the vast near zero eigenvalues as shown in Eq.(23).

For the empirical hessian $\tilde{H}_{L(\theta)} \triangleq \nabla_\theta \otimes \nabla_\theta \tilde{L}_J(\theta)$

$$\tilde{H}_{L(\theta)} = \frac{1}{J}\sum_{j=1}^{J}\left(\left[\frac{\partial \Upsilon_\theta(x_j)}{\partial \theta}\right]^T \cdot \left[\frac{\partial \Upsilon_\theta(x_j)}{\partial \theta}\right] + (\Upsilon_\theta(x_j) - y(x_j))\nabla_\theta \otimes \nabla_\theta \Upsilon_\theta(x_j)\right).$$

Due to the Weyl's inequality, we have

$$\lambda_k(H_{L(\theta)}) + \lambda_{\min}(\tilde{H}_{L(\theta)} - H_{L(\theta)}) \le \lambda_k(\tilde{H}_{L(\theta)}) \le \lambda_k(H_{L(\theta)}) + \lambda_{\max}(\tilde{H}_{L(\theta)} - H_{L(\theta)}) \tag{41}$$

We have

$$\lambda_{\max}(\tilde{H}_{L(\theta)} - H_{L(\theta)}) \le \left\|\tilde{H}_{L(\theta)} - H_{L(\theta)}\right\|_F \le \frac{C}{\sqrt{J}}(3L)^2$$

-> Not so useful....

Assume, error of the sample covariance? Gupta and Nagar, Matrix Variate Distributions, 2000)

$$\lambda_{\max}(\tilde{H}_{L(\theta)} - H_{L(\theta)}) \le$$

$$\mathbb{E}\left(\left\|\tilde{H}_{L(\theta)} - H_{L(\theta)}\right\|_F^2\right) = \frac{1}{J}\left(\mathrm{tr}(H_{L(\theta)}^2) + \mathrm{tr}^2(H_{L(\theta)})\right)$$

$$\left\|\tilde{H}_{L(\theta)} - H_{L(\theta)}\right\|_F \sim \frac{1}{\sqrt{J}}\sqrt{\mathrm{tr}(H_{L(\theta)}^2) + \mathrm{tr}^2(H_{L(\theta)})}$$

Still not a very tight bound.

3.2 Effective Dimension of Base Function

Define

$$H_\Upsilon = \int_\Omega \left[\sigma(w_l x + b_l)\right]_{l=1\cdots L} \left[\sigma(w_l x + b_l)\right]^T_{l=1\cdots L} \rho(x)\,\mathrm{d}x,$$

and

$$\tilde{H}_\Upsilon = \sum_{j=1}^{J} \left[\sigma(w_l x_j + b_l)\right]_{l=1\cdots L} \left[\sigma(w_l x_j + b_l)\right]^T_{l=1\cdots L}.$$

It is clear that H_Υ has the same eigenvalue decaying property as $H_{c(\theta)}$ or $H_{L(\theta)}$. And $\tilde{H}_\Upsilon$.... Ideally, the parameters $[a_l]_{l=1\cdots L}$ can be solved through

$$\begin{bmatrix} \sigma(w_1x_1+b_1) & \sigma(w_2x_1+b_2) & \cdots & \sigma(w_Lx_1+b_L) \\ \sigma(w_1x_2+b_1) & \sigma(w_2x_2+b_2) & \cdots & \sigma(w_Lx_2+b_L) \\ \vdots & & & \vdots \\ \sigma(w_1x_J+b_1) & \sigma(w_2x_J+b_2) & \cdots & \sigma(w_Lx_J+b_L) \end{bmatrix} \begin{bmatrix} a_1 \\ a_2 \\ \vdots \\ a_L \end{bmatrix} = \begin{bmatrix} y(x_1) \\ y(x_2) \\ \vdots \\ y(x_J) \end{bmatrix}$$

by its normal equation

$$\tilde{H}_\Upsilon \begin{bmatrix} a_1 \\ a_2 \\ \vdots \\ a_L \end{bmatrix} = \sum_{j=1}^{J} \left[\sigma(w_l x_j + b_l)\right]_{l=1\cdots L} y(x_j)$$

However, due to the analysis above $\tilde{H}_\Upsilon$ is numerically not full rank. To overcome this difficulty, one possibility is to design w_l and b_l so that $\tilde{H}_\Upsilon$ is reasonable full rank. But this could?? break the generalization of $\Upsilon_\theta(x)$?

References

[1] D. Gottlieb and S. A. Orszag, *Numerical analysis of spectral methods: theory and applications*, vol. 26, Siam, 1977.

[2] L. N. Trefethen, *Approximation Theory and Approximation Practice*, SIAM, Jan. 2013.